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### Signal transmission device, industrial apparatus and vehicle

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#### Abstract

The present disclosure provides a signal transmission device. The signal transmission device includes a transmitting circuit, a plurality of insulated transmission paths and a receiving circuit. The transmitting circuit is configured to generate a plurality of mixed data strings from a plurality of input data strings. The plurality of insulated transmission paths are configured for insulation transmission of the plurality of mixed data strings. The receiving circuit is configured to restore a plurality of output data strings corresponding to the plurality of input data strings from the plurality of mixed data strings that are insulated and transmitted via the plurality of insulated transmission paths.

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## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name | U.S. Cl. | CPC         |
|--------------|-------------|---------------|----------|-------------|
| 2010/0052826 | 12/2009     | Callahan      | 333/24R  | H01L 23/66  |
| 2015/0171901 | 12/2014     | Dupuis        | 375/295  | H04B 10/802 |

### FOREIGN PATENT DOCUMENTS

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|-------------|------------------|---------|-----|
| 2017-188903 | 12/2016          | JP      | N/A |

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## Background/Summary

### TECHNICAL FIELD

(1) The present disclosure relates to a signal transmission device, and an industrial apparatus and a vehicle using the same.

### BACKGROUND

(2) Conventionally, a signal transmission device that insulates between an input and an output and transmits signals is used in various application processes (power supply devices or motor drive devices).

(3) Moreover, an example in the related prior art may be referred from patent publication 1.

### PRIOR ART DOCUMENT

Patent Publication

(4) [Patent document 1] Japan Patent Publication No. 2017-188903

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram of a fundamental configuration of a signal transmission device.

(2) FIG. 2 is a diagram of a fundamental structure of a transformer chip.

- (3) FIG. 3 is a perspective diagram of a semiconductor device used as a two-channel transformer chip.
- (4) FIG. 4 is a plan view of the semiconductor device in FIG. 3.
- (5) FIG. 5 is a plan view of a layer in which is a low potential coil is formed in the semiconductor device in FIG. 3.
- (6) FIG. 6 is a plan view of a layer in which is a high potential coil is formed in the semiconductor device in FIG. 3.
- (7) FIG. 7 is a section diagram along a section line VIII-VIII in FIG. 6.
- (8) FIG. 8 is an enlarged diagram (a separation structure) of a region XIII shown in FIG. 7.
- (9) FIG. 9 is a diagram of a layout example of a transformer chip.
- (10) FIG. 10 is a diagram of a brief configuration of a signal transmission device.
- (11) FIG. 11 is a diagram of a first comparison example of a signal transmission device.
- (12) FIG. 12 is a diagram of a first comparison example (in malfunction) of a signal transmission device.
- (13) FIG. 13 is a diagram of a second comparison example of a signal transmission device.
- (14) FIG. 14 is a diagram of a signal transmission device according to an embodiment.
- (15) FIG. 15 is a diagram of a signal transmission device (in malfunction) according to an embodiment.
- (16) FIG. 16 is a diagram of a first example of a switching timing flag.
- (17) FIG. 17 is a diagram of a configuration example of a receiving circuit.
- (18) FIG. 18 is a diagram of an example of a signal transmission operation.
- (19) FIG. 19 is a diagram of a second example of a switching timing flag.
- (20) FIG. 20 is a diagram of a variation example of a flag adding operation.
- (21) FIG. 21 is a diagram of a third example of a switching timing flag.

#### DETAILED DESCRIPTION OF THE EMBODIMENTS

(22) <Signal Transmission Device (Fundamental Configuration)>

(23) FIG. 1 shows a diagram of a fundamental configuration of a signal transmission device. The signal transmission device **200** of this configuration example is a semiconductor integrated circuit device (a so-called gate-insulated driver IC) which insulates between a primary circuit system **200p** (a VCC1-GND1 system) and a secondary circuit system **200s** (a VCC2-GND2 system), transmits pulse signals from the primary circuit system **200p** to the secondary circuit system **200s**, and drives a gate of a switching element (not shown) disposed at the secondary circuit system **200s**. For example, the signal transmission device **200** seals a controller chip **210**, a driver chip **220** and a transformer chip **230** in one single package.

(24) The controller chip **210** is a semiconductor chip which receives a supply of a power supply voltage VCC1 (for example, using GND1 as a reference and being 7 V in maximum) and operates accordingly. For example, a pulse transmitting circuit **211** and buffers **212** and **213** are integrated in the controller chip **210**.

(25) The pulse transmitting circuit **211** is a pulse generator which generates transmission pulse signals **S11** and **S21** according to an input pulse signal IN. More specifically, the pulse transmitting circuit **211** performs pulse driving (an output of single or multiple transmitting pulses) of the transmission pulse signal **S11** when notified of a subject that the input pulse signal IN is at a high level, and performs pulse driving of the transmission pulse signal **S21** when notified of a subject that the input pulse signal IN is at a low level. That is to say, the pulse transmitting circuit **211** performs pulse driving on either of the transmission pulse signals **S11** and **S21** according to a logic level of the input pulse signal IN.

(26) The buffer **212** receives the input of the transmission pulse signal **S11** from the pulse transmitting circuit **211**, and performs pulse driving on the transformer chip **230** (more specifically, a transformer **231**).

(27) The buffer **213** receives the input of the transmission pulse signal **S21** from the pulse

transmitting circuit **211**, and performs pulse driving on the transformer chip **230** (more specifically, a transformer **232**).

(28) The driver chip **220** is a semiconductor chip which receives a supply of a power supply voltage VCC2 (for example, using GND2 as a reference and being 30 V in maximum) and operates accordingly. For example, buffers **221** and **222**, a pulse receiving circuit **223** and a driver **224** are integrated in the driver chip **220**.

(29) The buffer **221** performs waveform shaping on and outputs a sensed receiving pulse signal **S12** in the transformer chip **230** (more specifically, the transformer **231**) to the pulse receiving circuit **223**.

(30) The buffer **222** performs waveform shaping on and outputs a sensed receiving pulse signal **S22** in the transformer chip **230** (more specifically, the transformer **232**) to the pulse receiving circuit **223**.

(31) The pulse receiving circuit **223** drives the driver **224** according to reception pulse signals **S12** and **S22** input by the buffers **221** and **222** and generates an output pulse signal OUT. More specifically, the pulse receiving circuit **223** drives the driver **224** by receiving the pulse driving of the reception pulse signal **S12** to increase the output pulse signal OUT to a high level, while receiving the pulse driving of the reception pulse signal **S22** to decrease the output pulse signal OUT to a low level. That is to say, the pulse receiving circuit **223** switches the logic level of the output pulse signal OUT according to the logic level of the input pulse signal IN. Moreover, an RS trigger can be appropriately used as the pulse receiving circuit **223**.

(32) The driver **224** generates the output pulse signal OUT based on the driving control of the pulse receiving circuit **223**.

(33) The transformer chip **230** uses the transformers **231** and **232** as a DC insulation between the controller chip **210** and the driver chip **220**, and outputs the transmission pulse signals **S11** and **S21** input from the pulse transmitting circuit **211** as the reception pulse signals **S12** and **S22** to the pulse receiving circuit **223**. Moreover, in the present application, the term “DC insulation” mean that targets to be insulated are not connected by a conductor.

(34) More specifically, the transformer **231** outputs the reception pulse signal **S12** from a secondary-side coil **231s** according to the transmission pulse signal **S11** input to a primary-side coil **231p**. On the other hand, the transformer **232** outputs the reception pulse signal **S22** from a secondary-side coil **232s** according to the transmission pulse signal **S21** input to a primary-side coil **232p**.

(35) As such, in characteristics of a spiral coil used for insulation intercommunication, the input pulse signal IN is separated into the two transmission pulse signals **S11** and **S21** (equivalent to an amplified signal and an attenuated signal), and then transmitted from the primary circuit system **200p** to the secondary circuit system **200s** through two transformers **231** and **232**.

(36) Moreover, in addition to the controller chip **210** and the driver chip **220**, the signal transmission device **200** of this configuration example further includes the separate transformer **230** mounted with only the transformers **231** and **232**, and is formed in one single package by sealing these three chips.

(37) With the configuration above, since the controller chip **210** and the driver chip **220** can be formed by a common low withstand voltage to a medium withstand voltage (a withstand voltage of several V to tens of V) processes, manufacturing costs can be reduced without use of specified high withstand voltage technique (a withstand voltage of several kV).

(38) Moreover, for example, the signal transmission device **200** can be appropriately used for power supply devices or motor drive devices mounted on vehicles. In addition to an engine, the vehicle further includes an xEV such as a battery electric vehicle (BEV), a hybrid electric vehicle (HEV), a plug-in hybrid electric vehicle/plug-in hybrid vehicle (PHEV/PHV), or a fuel cell electric vehicle/fuel cell vehicle (FCEV/FCV).

(39) <Transformer Chip (Fundamental Structure)>

(40) Next, the fundamental structure of the transformer chip **230** is described below. FIG. 2 shows a diagram of a fundamental structure of the transformer chip **230**. In the transformer chip **230** of the drawing, the transformer **231** includes a primary-side coil **231p** and a secondary-side coil **231s** facing each other in a top-down direction. The transformer **232** includes a primary-side coil **232p** and a secondary-side coil **232s** facing each other in a top-down direction.

(41) Both of the primary-side coils **231p** and **232p** are formed in a first wiring layer (a lower layer) **230a** of the transformer chip **230**. Both of the secondary-side coils **231s** and **232s** are formed in a second wiring layer (an upper layer in the drawing) **230b** of the transformer chip **230**. Moreover, the secondary-side coil **231s** is disposed directly above the primary-side coil **231p**, and faces the primary-side coil **231p**. In addition, the secondary-side coil **232s** is disposed directly above the primary-side coil **232p**, and faces the primary-side coil **232p**.

(42) The primary-side coil **231p** is laid out in a spiral shape starting from a first end connected to an internal terminal X21 so as to surround the internal terminal X21 clockwise, and a second end equivalent to an end is connected to an internal terminal X22. On the other hand, the primary-side coil **232p** is laid out in a spiral shape starting from a first end connected to an internal terminal X23 so as to surround the internal terminal X23 counterclockwise, and a second end equivalent to an end is connected to the internal terminal X22. The internal terminals X21, X22 and X23 are linearly arranged according to an order shown in the drawing.

(43) The internal terminal X21 is connected to an external terminal T21 of the second layer **230b** via a conductive wiring Y21 and a path Z21. The internal terminal X22 is connected to an external terminal T22 of the second layer **230b** via a conductive wiring Y22 and a path Z22. The internal terminal X23 is connected to an external terminal T23 of the second layer **230b** via a conductive wiring Y23 and a path Z23. Moreover, the external terminals T21 to T23 are linearly arranged and are configured to bond with conductive wires of the controller chip **210**.

(44) The secondary-side coil **231s** is laid out in a spiral shape starting from a first end connected to an external terminal T24 so as to surround the external terminal T24 counterclockwise, and a second end equivalent to an end is connected to an external terminal T25. On the other hand, the secondary-side coil **232s** is laid out in a spiral shape starting from a first end connected to an external terminal T26 so as to surround the external terminal T26 clockwise, and a second end equivalent to an end is connected to the external terminal T25. Moreover, the external terminals T24, T25 and T26 are linearly arranged in an order shown in the drawing, and are configured to bond with conductive wires of the driver chip **220**.

(45) The secondary-side coils **231s** and **232s** are respectively alternating-current (AC) connected to the primary-side coils **231p** and **232p** by magnetic coupling, and are DC insulated from the primary-side coils **231p** and **232p**. That is to say, the driver chip **220** is AC connected to the controller chip **210** via the transformer chip **230**, and is DC insulated from the controller chip **210** via the transformer chip **230**.

(46) <Transformer Chip (Two-Channel Type)>

(47) FIG. 3 shows a perspective diagram of a semiconductor device **5** used as a two-channel transformer chip. FIG. 4 shows a plan view of the semiconductor device **5** in FIG. 3. FIG. 5 shows a plan view of a layer in which a low potential coil **22** (equivalent to a primary-side coil of a transformer) is formed in the semiconductor device **5** in FIG. 3. FIG. 6 shows a plan view of a layer in which a high potential coil **23** (equivalent to a secondary-side coil of a transformer) is formed in the semiconductor device **5** in FIG. 3. FIG. 7 shows a section diagram along a section line VIII-VIII in FIG. 6. FIG. 8 shows an enlarged diagram of a region XIII shown in FIG. 7 and represents a diagram of a separation structure **130**.

(48) Referring to FIG. 3 to FIG. 7, the semiconductor device **5** includes a semiconductor chip **41** formed to have a rectangular shape. The semiconductor chip **41** includes at least one of silicon, a wide bandgap semiconductor and a compound semiconductor.

(49) The wide bandgap semiconductor includes a semiconductor having a bandgap greater than that

of silicon (about 1.12 eV). The bandgap of the wide bandgap semiconductor is preferably 2.0 eV or more. The wide bandgap semiconductor can be silicon carbide (SiC). The compound semiconductor can be a III-V group compound semiconductor. The compound semiconductor can include at least one of aluminum nitride (AlN), indium nitride (InN), gallium nitride (GaN), and gallium arsenide (GaAs).

(50) The semiconductor chip **41** includes a semiconductor substrate formed of silicon in this implementation. The semiconductor chip **41** can also be an epitaxial substrate having a laminated structure including a silicon semiconductor substrate and a silicon epitaxial layer. A conductivity type of the semiconductor substrate can be n type or p type. The epitaxial layer can be n type or p type.

(51) The semiconductor chip **41** includes a first main surface **43** on one side, a second main surface **44** on the other side, and four chip sidewalls **44A** to **44D** connecting the first main surface **42** and the second main surface **43**. The first main surface **42** and the second main surface **43** are formed to have quadrilateral shapes (specifically rectangular shapes in this implementation) in a plan view when viewing from a normal direction Z thereof (to be referred to as “in the plan view” below).

(52) The chip sidewalls **44A** to **44D** include a first chip sidewall **44A**, a second chip sidewall **44B**, a third chip sidewall **44C** and a fourth chip sidewall **44D**. The first chip sidewall **44A** and the second chip sidewall **44B** form long sides of the semiconductor chip **41**. The first chip sidewall **44A** and the second chip sidewall **44B** extend in a first direction X and are opposite in a second direction Y. The third chip sidewall **44C** and the fourth chip sidewall **44D** form short sides of the semiconductor chip **41**. The third chip sidewall **44C** and the fourth chip sidewall **44D** extend in the second direction Y and are opposite in the first direction X. The chip sidewalls **44A** and **44D** include grinding surfaces.

(53) The semiconductor device **5** further includes an insulating layer **51** formed on the first main surface **42** of the semiconductor chip **41**. The insulating layer **51** has an insulating main surface **52** and insulating sidewalls **53A** to **53D**. The insulating main surface **52** is formed to have a quadrilateral shape (a rectangular shape in this implementation) matching the first main surface **42**. The insulating main surface **52** extends in parallel with respect to the first main surface **42**.

(54) The insulating sidewalls **53A** to **53D** include a first insulating sidewall **53A**, a second insulating sidewall **53B**, a third insulating sidewall **53C** and a fourth insulating sidewall **53D**. The insulating sidewalls **53A** to **53D** extend from a periphery of the insulating main surface **52** toward the semiconductor chip **41**, and are connected to the chip sidewalls **44A** to **44D**. More specifically, the insulating sidewalls **53A** to **53D** are formed as leveled surfaces with respect to the chip sidewalls **44A** to **44D**. The insulating sidewalls **53A** to **53D** are formed as leveled grinding surfaces with respect to the chip sidewalls **44A** to **44D**.

(55) The insulating layer **51** is formed by a laminated structure including a lowermost insulating layer **55**, an uppermost insulating layer **56** and multiple (**11** in this embodiment) interlayer insulating layers **57**. The lowermost insulating layer **55** is an insulating layer directly covering the first main surface **42**. The uppermost insulating layer **56** is an insulating layer forming the insulating main surface **52**. The multiple interlayer insulating layers **57** are insulating layers located between the lowermost insulating layer **55** and the uppermost insulating layer **56**. The lowermost insulating layer **55** includes a single-layer structure containing SiO in this implementation. The uppermost insulating layer **56** includes a single-layer structure containing SiO in this implementation. Each of a thickness of the lowermost insulating layer **55** and a thickness of the uppermost insulating layer **56** can be between about 1 micrometer ( $\mu\text{m}$ ) and about 3  $\mu\text{m}$  (for example, approximately 2  $\mu\text{m}$ ).

(56) Each of the multiple interlayer insulating layers **57** includes a laminated structure including a first insulating layer **58** on the side of the lowermost insulating layer **55** and a second insulating layer **59** on the side of the uppermost insulating layer **56**. The first insulating layer **58** can include silicon nitride (SiN). The first insulating layer **58** is formed of an etch barrier layer for the second

insulating layer **59**. A thickness of the first insulating layer **58** can be between about 0.1  $\mu\text{m}$  and about 1  $\mu\text{m}$  (for example, approximately 0.3  $\mu\text{m}$ ).

(57) The second insulating layer **59** is formed on the first insulating layer **58**, and includes an insulating material different from that of the first insulating layer **58**. The second insulating layer **59** can include SiO. A thickness of the second insulating layer **59** can be between about 1  $\mu\text{m}$  and about 3  $\mu\text{m}$  (for example, approximately 2  $\mu\text{m}$ ). The thickness of the second insulating layer **59** is preferably greater than the thickness of the first insulating layer **58**.

(58) A total thickness DT of the insulating layer **51** is between about 5  $\mu\text{m}$  and about 50  $\mu\text{m}$ . The total thickness DT of the insulating layer **51** and the number of the interlayer insulating layers **57** laminated can be any as desired, and are adjusted according to an insulation withstand voltage (an insulation breakdown voltage) to be implemented. Moreover, insulating materials of the lowermost insulating layer **55**, the uppermost insulating layer **56** and the interlayer insulating layers **57** can be any as desired, and are not limited to specific insulating materials.

(59) The semiconductor device **5** includes a first functional component **45** formed at the insulating layer **51**. The first functional component **45** includes one or more (multiple in this implementation) transformers **21** (equivalent to the transformer above). That is to say, the semiconductor device **5** is a multi-channel component including multiple transformers **21**. The multiple transformers **21** are formed at a distance from the insulating sidewalls **53A** to **53D** on an inside of the insulating layer **51**. The multiple transformers **21** are formed at intervals in the first direction X.

(60) More specifically, the multiple transformers **21** include a first transformer **21A**, a second transformer **21B**, a third transformer **21C** and a fourth transformer **21D** sequentially formed from the side of the insulating sidewall **53C** toward the side of the insulating sidewall **53D** in the plan view. Each of the multiple transformers **21A** to **21D** has a structure identical to that of another. The structure of the first transformer **21A** is described as an example below. Details of the structure of the first transformer **21A** are applicable to the structures of the second transformer **21B**, the third transformer **21C** and the fourth transformer **21D**, and such repeated description is omitted herein.

(61) Referring to FIG. 5 to FIG. 7 the first transformer **21A** includes a low potential coil **22** and a high potential coil **23**. The low potential coil **22** is formed in the insulating layer **51**. The high potential coil **23** is formed in the insulating layer **51** to face the low potential coil **22** in the normal direction Z. The low potential coil **22** and the high potential coil **23** are in a region sandwiched by the lowermost insulating layer **55** and the uppermost insulating layer **56** in this implementation (that is, the multiple interlayer insulating layers **57**).

(62) The low potential coil **22** is formed on the side of the lowermost insulating layer **55** (the semiconductor chip **41**) in the insulating layer **51**, and the high potential coil **23** is formed on the side of the uppermost insulating layer **56** with respect to the low potential coil **22** in the insulating layer **51**. That is to say, the high potential coil **23** is separated by the low potential coil **22** to face the semiconductor chip **41**. The low potential coil **22** and the high potential coil **23** can be arranged at any positions desired. Moreover, the high potential coil **23** is separately by one or more interlayer insulating layers **57** to face the low potential coil **22**.

(63) A distance between the low potential coil **22** and the high potential coil **23** (that is, the number of the interlayer insulating layers **57** laminated) is appropriately adjusted according to the insulation withstand voltage and an electric field strength between the low potential coil **22** and the high potential coil **23**. In this implementation, the low potential coil **22** is formed at the third interlayer insulating layer **57** from the side of the lowermost insulating layer **55**. In this implementation, the high potential coil **23** is formed at the first interlayer insulating layer **57** from the side of the uppermost insulating layer **56**.

(64) The low potential coil **22** is buried by passing through the first insulating layer **58** and the second insulating layer **59** in the interlayer insulating layers **57**. The low potential coil **22** includes a first inner end **24**, a first outer end **25**, and a first spiral portion **26** wound spirally between the first inner end **24** and the first outer end **25**. The first spiral portion **26** winds as a spiral extending

in an ellipsoidal shape (a long circular shape). An innermost peripheral portion forming the first spiral portion **26** defines a first inner region **66** in an ellipsoidal shape in the plan view.

(65) The number of windings of the first spiral portion **26** can be between 5 and 30. A width of the first spiral portion **26** can be between about 0.1  $\mu\text{m}$  and about 5  $\mu\text{m}$ . The width of the first spiral portion **26** is preferably between about 1  $\mu\text{m}$  and about 3  $\mu\text{m}$ . The width of the first spiral portion **26** is defined by a width in a direction perpendicular to the spiral direction. A first winding pitch of the first spiral portion **26** can be between about 0.1  $\mu\text{m}$  and about 5  $\mu\text{m}$ . The first winding pitch is preferably between about 1  $\mu\text{m}$  and about 3  $\mu\text{m}$ . The first winding pitch is defined by a distance between two adjacent portions of the first winding portion **26** in a direction perpendicular to the spiral direction.

(66) A winding shape of the first spiral portion **26** and a planar shape of the first inner region **66** can be any as desired, and are not limited to the examples shown in the implementation in FIG. 5. The first spiral portion **26** can also be wound in a shape as a polygon such as a triangle or a quadrilateral or a circle in the plan view. The first inner region **66** can also be defined in a shape as a polygon such as a triangle or a quadrilateral or a circle in the plan view according to the winding shape of the first spiral portion **26**.

(67) The low potential coil **22** can include at least one of titanium (Ti), titanium nitride (TiN), copper (Cu), aluminum (Al) and tungsten (W). The low potential coil **22** can also have a laminated structure including a barrier layer and a body layer. The barrier layer defines a recess space in the interlayer insulating layer **57**. The barrier layer can include at least one of Ti and TiN. The body layer can include at least one of Cu, Al and W.

(68) The high potential coil **23** is buried by passing through the first insulating layer **58** and the second insulating layer **59** in the interlayer insulating layers **57**. The high potential coil **23** includes a second inner end **27**, a second outer end **28**, and a second spiral portion **29** wound spirally between the second inner end **27** and the second outer end **28**. The second spiral portion **29** winds as a spiral extending in an ellipsoidal shape (a long circular shape) in the plan view. In this implementation, an innermost peripheral portion forming the second spiral portion **29** defines a second inner region **67** in an ellipsoidal shape in the plan view. The second inner region **67** of the second spiral portion **29** faces the first inner region **66** of the first spiral portion **26** in the normal direction Z.

(69) The number of windings of the second spiral portion **29** can be between 5 and 30. The number of windings of the second spiral portion **29** with respect to the number of windings of the first spiral portion **26** is adjusted according to a voltage value to be stepped up. The number of windings of the second spiral portion **29** is preferably greater than the number of windings of the first spiral portion **26**. As a matter of course, the number of windings of the second spiral portion **29** can be less than the number of windings of the first spiral portion **26** or can be equal to the number of windings of the first spiral portion **26**.

(70) A width of the second spiral portion **29** can be between about 0.1  $\mu\text{m}$  and about 5  $\mu\text{m}$ . The width of the second spiral portion **29** is preferably between about 1  $\mu\text{m}$  and about 3  $\mu\text{m}$ . The width of the second spiral portion **29** is defined by a width in a direction perpendicular to the spiral direction. The width of the second spiral portion **29** is preferably equal to the width of the first spiral portion **26**.

(71) A second winding pitch of the second spiral portion **29** can be between about 0.1  $\mu\text{m}$  and about 5  $\mu\text{m}$ . The second winding pitch is preferably between about 1  $\mu\text{m}$  and about 3  $\mu\text{m}$ . The second winding pitch is defined by a distance between two adjacent portions of the second winding portion **29** in a direction perpendicular to the spiral direction. The second winding pitch is preferably equal to the first winding pitch of the first spiral portion **26**.

(72) A winding shape of the second spiral portion **29** and a planar shape of the second inner region **67** can be any as desired, and are not limited to the examples shown in the implementation in FIG. 6. The second spiral portion **29** can also be wound in a shape as a polygon such as a triangle or a



quadrilateral or a circle in the plan view. The second inner region **67** can also be defined to have a shape as a polygon such as a triangle or a quadrilateral or a circle in the plan view according to the winding shape of the second spiral portion **29**.

(73) The high potential coil **23** is preferably formed of a same conductive material as the low potential coil **22**. That is to say, the high potential coil **23** preferably includes a barrier layer and a body layer as the low potential coil **22**.

(74) Referring to FIG. **4**, the semiconductor device **5** includes multiple (**12** in this drawing) low potential terminals **11** and multiple (**12** in this drawing) high potential terminals **12**. The multiple low potential terminals **11** are electrically connected to the low potential coils **22** of the corresponding transformers **21A** to **21D**, respectively. The multiple high potential terminals **12** are electrically connected to the high potential coils **23** of the corresponding transformers **21A** to **21D**, respectively.

(75) The multiple low potential terminals **11** are formed on the insulating main surface **52** of the insulating layer **51**. More specifically, the multiple low potential terminals **11** are formed at a distance from the multiple transformers **21A** to **21D** in the second direction **Y** in a region on the side of the insulating sidewall **53B**, and are arranged at intervals in the first direction **X**.

(76) The multiple low potential terminals **11** include a first low potential terminal **11A**, a second low potential terminal **11B**, a third low potential terminal **11C**, a fourth low potential terminal **11D**, a fifth low potential terminal **11E** and a sixth low potential terminal **11F**. In this implementation, each of the low potential terminals **11A** to **11F** is formed as two in number. The number of the low potential terminals **11A** to **11F** can be any as desired.

(77) The first low potential terminal **11A** faces the first transformer **21A** in the second direction **Y** in the plan view. The second low potential terminal **11B** faces the second transformer **21B** in the second direction **Y** in the plan view. The third low potential terminal **11C** faces the third transformer **21C** in the second direction **Y** in the plan view. The fourth low potential terminal **11D** faces the fourth transformer **21D** in the second direction **Y** in the plan view. The fifth low potential terminal **11E** is formed in a region between the first low potential terminal **11A** and the second low potential terminal **11B** in the plan view. The sixth low potential terminal **11F** is formed in a region between the third low potential terminal **11C** and the fourth low potential terminal **11D** in the plan view.

(78) The first low potential terminal **11A** is electrically connected to the first inner end **24** of the first transformer **21A** (the low potential coil **22**). The second low potential terminal **11B** is electrically connected to the first inner end **24** of the second transformer **21B** (the low potential coil **22**). The third low potential terminal **11C** is electrically connected to the first inner end **24** of the third transformer **21C** (the low potential coil **22**). The fourth low potential terminal **11D** is electrically connected to the first inner end **24** of the fourth transformer **21D** (the low potential coil **22**).

(79) The fifth low potential terminal **11E** is electrically connected to the first outer end **25** of the first transformer **21A** (the low potential coil **22**) and the first outer end **25** of the second transformer **21B** (the low potential coil **22**). The sixth low potential terminal **11F** is electrically connected to the first outer end **25** of the third transformer **21C** (the low potential coil **22**) and the first outer end **25** of the fourth transformer **21D** (the low potential coil **22**).

(80) The multiple high potential terminals **11** are formed at a distance from the multiple low potential terminals **11** on the insulating main surface **52** of the insulating layer **51**. More specifically, the multiple high potential terminals **12** are formed at a distance from the multiple low potential terminals **11** in the second direction **Y** in a region on the side of the insulating sidewall **53A**, and are arranged at intervals in the first direction **X**.

(81) The multiple high potential terminals **12** are formed in regions close to the corresponding transformers **21A** to **21D** in the plan view, respectively. The high potential terminals **12** being close to the transformers **21A** to **21D** mean that, a distance between the high potential terminals **21** and

the transformers **21** is less than a distance between the low potential terminals **11** and the high potential terminals **12** in the plan view.

(82) More specifically, the multiple high potential terminals **12** are formed at intervals in the first direction X to face the multiple transformers **21A** to **21D** in the first direction X in the plan view. More specifically, the multiple high potential terminals **12** are formed at intervals in the first direction X in a region located between the second inner region **67** of the high potential coils **23** and the adjacent high potential coils **23** in the plan view. Accordingly, the multiple high potential terminals **12** are formed in one row with the multiple transformers **21A** to **21D** in the first direction X in the plan view.

(83) The multiple high potential terminals **12** include a first high potential terminal **12A**, a second high potential terminal **12B**, a third high potential terminal **12C**, a fourth high potential terminal **12D**, a fifth high potential terminal **12E** and a sixth high potential terminal **12F**. In this implementation, each of the high potential terminals **12A** to **12F** is formed as two in number. The number of the high potential terminals **12A** to **12F** can be any as desired.

(84) The first high potential terminal **12A** is formed in the second inner region **67** of the first transformer **21A** (the high potential coil **23**) in the plan view. The second high potential terminal **12B** is formed in the second inner region **67** of the second transformer **21B** (the high potential coil **23**) in the plan view. The third high potential terminal **12C** is formed in the second inner region **67** of the third transformer **21C** (the high potential coil **23**) in the plan view. The fourth high potential terminal **12D** is formed in the second inner region **67** of the fourth transformer **21D** (the high potential coil **23**) in the plan view. The fifth high potential terminal **12E** is formed in a region between the first transformer **21A** and the second transformer **21B** in the plan view. The sixth high potential terminal **12F** is formed in a region between the third transformer **21C** and the fourth transformer **21D** in the plan view.

(85) The first high potential terminal **12A** is electrically connected to the second inner end **27** of the first transformer **21A** (the high potential coil **23**). The second high potential terminal **12B** is electrically connected to the second inner end **27** of the second transformer **21B** (the high potential coil **23**). The third high potential terminal **12C** is electrically connected to the second inner end **27** of the third transformer **21C** (the high potential coil **23**). The fourth high potential terminal **12D** is electrically connected to the second inner end **27** of the fourth transformer **21D** (the high potential coil **23**).

(86) The fifth high potential terminal **12E** is electrically connected to the second outer end **28** of the first transformer **21A** (the high potential coil **23**) and the second outer end **28** of the second transformer **21B** (the high potential coil **23**). The sixth high potential terminal **12F** is electrically connected to the second outer end **28** of the third transformer **21C** (the high potential coil **23**) and the second outer end **28** of the fourth transformer **21D** (the high potential coil **23**).

(87) Referring to FIG. 5 to FIG. 7, the semiconductor device **5** includes a first low potential wiring **31**, a second low potential wiring **32**, a first high potential wiring **33** and a second high potential wiring **34** individually formed in the first insulating layer **51**. In this implementation, a plurality of first low potential wirings **31**, a plurality of second low potential wirings **32**, a plurality of first high potential wirings **33** and a plurality of second high potential wirings **34** are formed.

(88) The first low potential wiring **31** and the second low potential wiring **32** fix the low potential coil **22** of the first transformer **21A** and the low potential coil **22** of the second transformer **21B** to the same potential. Moreover, the first low potential wiring **31** and the second low potential wiring **32** fix the low potential coil **22** of the third transformer **21C** and the low potential coil **22** of the fourth transformer **21D** to the same potential. In this implementation, the first low potential wiring **31** and the second low potential wiring **32** fix all of the low potential coils **22** of the transformers **21A** to **21D** to the same potential.

(89) The first high potential wiring **33** and the second high potential wiring **34** fix the high potential coil **23** of the first transformer **21A** and the high potential coil **23** of the second transformer **21B** to

the same potential. Moreover, the first high potential wiring **33** and the second high potential wiring **34** fix the high potential coil **23** of the third transformer **21C** and the high potential coil **23** of the fourth transformer **21D** to the same potential. In this implementation, the first high potential wiring **33** and the second high potential wiring **34** fix all of the high potential coils **23** of the transformers **21A** to **21D** to the same potential.

(90) The plurality of first low potential wirings **31** are electrically connected to the corresponding low potential terminals **11A** to **11D** and the first inner ends **24** of the corresponding transformers **21A** to **21D** (the low potential coils **22**), respectively. The plurality of first low potential wirings **31** have the same structure. The structure of the first low potential wiring **31** connected to the first low potential terminal **11A** and the first transformer **21A** is described as an example below. Details of the structure of the first low potential wiring **31** connected to the first transformer **21A** are applicable to the structures of other first low potential wirings **31**, and such repeated description is omitted herein.

(91) The first low potential wiring **31** includes a through wiring **71**, a low potential connection wiring **72**, a lead-out wiring **73**, a first connection plug electrode **74**, a second connection plug electrode **75**, one or more (multiple in this implementation) pad plug electrodes **76**, and one or more (multiple in this implementation) substrate plug electrodes **77**.

(92) Each of the through wiring **71**, the low potential connection wiring **72**, the lead-out wiring **73**, the first connection plug electrode **74**, the second connection plug electrode **75**, the pad plug electrode **76** and the substrate plug electrode **77** is preferably formed of the same conductive material as the low potential coil **22**. That is to say, each of the through wiring **71**, the low potential connection wiring **72**, the lead-out wiring **73**, the first connection plug electrode **74**, the second connection plug electrode **75**, the pad plug electrode **76** and the substrate plug electrode **77** preferably includes a barrier layer and a body layer, as the low potential coil **22**.

(93) The through wiring **71** passes through the plurality of interlayer insulating layers **57** in the insulating layer **51**, and extends in a column shape along the normal direction Z. In this implementation, the through wiring **71** is formed in a region between the lowermost insulating layer **55** and the uppermost insulating layer **56** in the insulating layer **51**. The through wiring **71** includes an upper end on the side of the uppermost insulating layer **56** and a lower end on the side of the lowermost insulating layer **55**. The upper end of the through wiring **71** is formed in the same interlayer insulating layer **57** as the high potential coil **23**, and is covered by the uppermost insulating layer **56**. The upper end of the through wiring **71** is formed in the same interlayer insulating layer **57** as the low potential coil **22**.

(94) In this implementation, the through wiring **71** includes a first electrode layer **78**, a second electrode layer **79** and a plurality of wiring plug electrodes **80**. In the through wiring **71**, each of the first electrode layer **78**, the second electrode layer **79** and the wiring plug electrodes **80** is formed of the same conductive material as the low potential coil **22**. That is to say, each of the first electrode layer **78**, the second electrode layer **79** and the wiring plug electrodes **80** includes a barrier layer and a body layer, as the potential coil **22**.

(95) The first electrode layer **78** forms the upper end of the through wiring **71**. The second electrode layer **79** forms the lower end of the through wiring **71**. The first electrode layer **78** is formed to have a shape of an island, and faces the low potential terminal **11** (the first low potential terminal **11A**) in the normal direction Z. The second electrode layer **79** is formed to have a shape of an island, and faces the first electrode layer **78** in the normal direction Z.

(96) The plurality of wiring plug electrodes **80** are respectively buried in the plurality of interlayer insulating layers **57** in a region between the first electrode layer **78** and the second electrode layer **79**. The plurality of wiring plug electrodes **80** are laminated from the lowermost insulating layer **55** toward the uppermost insulating layer **56** to be electrically connected to one another, and electrically connect the first electrode layer **78** and the second electrode layer **79**. Each of the plurality of wiring plug electrodes **80** has a planar area less than a planar area of the first electrode

layer **78** and a planar area of the second electrode layer **79**.

(97) Moreover, the number of the plurality of wiring plug electrodes **80** laminated is consistent with the number of the plurality of interlayer insulating layers **57** laminated. In this implementation, six wiring plug electrodes **80** are buried in the interlayer insulating layers **57**; however, the number of the wiring plug electrodes **80** buried in the interlayer insulating layers **57** can be any number as desired. As a matter of course, one or more wiring plug electrodes **80** passing through the plurality of interlayer insulating layers **57** can also be formed.

(98) The low potential connection wiring **72** is in the same interlayer insulating layer **57** as the low potential coil **22**, and is formed in the first inner region **66** of the first transformer **21A** (the low potential coil **22**). The low potential connection wiring **72** is formed to have a shape of an island, and faces the high potential terminal **12** (the first high potential terminal **12A**) in the normal direction Z. The low potential connection wiring **72** preferably has a planar area greater than the planar area of the wiring plug electrode **80**. The low potential connection wiring **72** is electrically connected to the first inner end **24** of the low potential coil **22**.

(99) The lead-out wiring **73** is formed in a region between the semiconductor chip **41** and the through wiring **71** in the interlayer insulating layer **57**. In this implementation, the lead-out wiring **73** is formed at the first interlayer insulating layer **57** from the lowermost insulating layer **55**. The lead-out wiring **73** includes a first end on one side, a second end on the other side, and a wiring portion connecting the first end and the second unit. The first end of the lead-out wiring **73** is located in a region between the semiconductor chip **41** and the lower end of the through wiring **71**. The second end of the lead-out wiring **73** is located in a region between the semiconductor chip **41** and the low potential connection wiring **72**. The wiring portion extends along the first main surface **42** of the semiconductor chip **41**, and extends as a strip in a region between the first end and the second end.

(100) The first connection plug electrode **74** is formed in a region between the through wiring **71** and the lead-out wiring **73** in the interlayer insulating layers **57**, and is electrically connected to the through wiring **71** and the first end of the lead-out wiring **73**. The second connection plug electrode **75** is formed in a region between the low potential connection wiring **72** and the lead-out wiring **73** in the interlayer insulating layers **57**, and is electrically connected to the low potential connection wiring **72** and the second end of the lead-out wiring **73**.

(101) The plurality of pad plug electrodes **76** are formed in a region between the low potential terminal **11** (the first low potential terminal **11A**) and the through wiring **71** in the uppermost insulating layer **56**, and are individually electrically connected to the low potential terminal **11** and the upper end of the through wiring **71**. The plurality of substrate plug electrodes **77** are formed in a region between the semiconductor chip **41** and the lead-out wiring **73** in the lowermost insulating layer **55**. In this implementation, the substrate plug electrodes **77** are formed in a region between the semiconductor chip **41** and the first end of the lead-out wiring **73**, and are individually electrically connected to the semiconductor chip **41** and the first end of the lead-out wiring **73**.

(102) Referring to FIG. 6 and FIG. 7, the plurality of first high potential wirings **33** are electrically connected to the corresponding high potential terminals **12A** to **12D** and the second inner ends **27** of the corresponding transformers **21A** to **21D** (the high potential coils **23**), respectively. The plurality of first high potential wirings **33** have the same structure. The structure of the first high potential wiring **33** connected to the first high potential terminal **12A** and the first transformer **21A** is described as an example below. Details of the structure of the first high potential wiring **33** connected to the first transformer **21A** are applicable to the structures of other first high potential wirings **33**, and such repeated description is omitted herein.

(103) The first high potential wiring **33** includes a high potential connection wiring **81**, and one or more (multiple in this implementation) pad plug electrodes **82**. The high potential connection wiring **81** and the pad plug electrode **82** are preferably formed of a same conductive material as the low potential coil **22**. That is to say, each of the high potential connection wiring **81** and the pad

plug electrode **82** preferably includes a barrier layer and a body layer, as the low potential coil **22**. (104) The high potential connection wiring **81** is in the same interlayer insulating layer **57** as the high potential coil **23**, and is formed in the second inner region **67** of the high potential coil **23**. The high potential connection wiring **81** is formed to have a shape of an island, and faces the high potential terminal **12** (the first high potential terminal **12A**) in the normal direction Z. The high potential connection wiring **81** is electrically connected to the second inner end **27** of the high potential coil **23**. The high potential connection wiring **81** is formed at a distance from the low potential connection wiring **72** in the plan view, and does not face the low potential connection wiring **72** in the normal direction Z. Accordingly, an insulation distance between the low potential connection wiring **72** and the high potential connection wiring **81** is increased, and so the insulation withstand voltage of the insulating layer **51** is increased.

(105) The plurality of pad plug electrodes **82** are formed in a region between the high potential terminal **12** (the first high potential terminal **12A**) and the high potential connection wiring **81** in the uppermost insulating layer **56**, and are individually electrically connected to the high potential terminal **12** and the high potential connection wiring **81**. Each of the plurality of pad plug electrodes **82** has a planar area less than a planar area of the high potential connection wiring **81**.

(106) Referring to FIG. 7, a distance **D1** between the low potential terminal **11** and the high potential terminal **12** is preferably greater than a distance **D2** between the low potential coil **22** and the high potential coil **23** ( $D2 < D1$ ). The distance **D1** is preferably greater than a total thickness **DT** of the plurality of interlayer insulating layers **57** ( $DT < D1$ ). A ratio  $D2/D1$  of the distance **D2** to the distance **D1** can be between about 0.01 and about 0.1. The distance **D1** is preferably between about 100  $\mu\text{m}$  and about 500  $\mu\text{m}$ . The distance **D2** can also be between about 1  $\mu\text{m}$  and about 50  $\mu\text{m}$ . The distance **D2** is preferably between about 5  $\mu\text{m}$  and about 25  $\mu\text{m}$ . Values of the distance **D1** and the distance **D2** can be any as desired, and are appropriately adjusted according to an insulation withstand voltage to be implemented.

(107) Referring to FIG. 6 and FIG. 7, the semiconductor device **7** includes a dummy pattern **85** buried in the insulating layer **51** to be located around the transformers **21A** to **21D** in the plan view.

(108) The dummy pattern **85** is formed by a pattern different (a discontinuous pattern) from the high potential coil **23** and the low potential coil **22**, and is independent from the transformers **21A** to **21D**. That is to say, the dummy pattern **75** does not function as the transformers **21A** to **21D**. The dummy pattern **85** is formed as a shielding conductor layer that shields an electric field between the low potential coils **22** and the high potential coils **23** in the transformers **21A** to **21D**, and suppresses the concentration of the electric field with respect to the high potential coil **23**. In this implementation, the dummy pattern **85** is laid out in a round-about manner with a linear density equal to a linear density of the high potential coil **23** per unit area. The linear density of the dummy pattern **85** being equal to the linear density of the high potential coil **23** means that the linear density of the dummy pattern falls within a range of +20% of the linear density of the high potential coil **23**.

(109) A depth position of the dummy pattern **85** inside the insulating layer **51** can be any as desired, and is adjusted according to an intensity of an electric field to be alleviated. The dummy pattern **85** is preferably in a region close to the high potential coil **23** with respect to the low potential coil **22** in the normal direction Z. Moreover, the dummy pattern **85** being close to the high potential coil **23** in the normal direction Z means that, in the normal direction Z, a distance between the dummy pattern **85** and the high potential coil **23** is less than a distance between the dummy pattern **85** and the low potential coil **22**.

(110) In this case, the concentration of the electric field with respect to the high potential coil **23** can be appropriately suppressed. In the normal direction Z, the concentration of the electric field with respect to the high potential coil **23** can be better suppressed as the distance between the dummy pattern **85** and the high potential coil **23** decreases. The dummy pattern **85** is preferably formed in the same interlayer insulating layer **57** as the high potential coil **23**. In this case, the

concentration of the electric field with respect to the high potential coil **23** can be even more appropriately suppressed. The dummy pattern **85** includes a plurality of dummy patterns with different electrical states. The dummy pattern **85** can also include a high potential dummy pattern **86**.

(111) A depth position of the high potential dummy pattern **86** inside the insulating layer **51** can be any as desired, and is adjusted according to an intensity of an electric field to be alleviated. The high potential dummy pattern **86** is preferably in a region close to the high potential coil **23** with respect to the low potential coil **22** in the normal direction Z. Moreover, the high potential dummy pattern **86** being close to the high potential coil **23** in the normal direction Z means that, in the normal direction Z, a distance between the high potential dummy pattern **86** and the high potential coil **23** is less than a distance between the high potential dummy pattern **86** and the low potential coil **22**.

(112) The dummy pattern **85** includes a floating dummy pattern formed in an electrically floating state in the insulating layer **51** so as to be located around the transformers **21A** to **21D** in the plan view.

(113) In this implementation, the floating dummy pattern is laid out in a round-about manner in a tight line to partially cover and partially expose a region around the high potential coil **23** in the plan view. The floating dummy pattern can be formed to have a shape of with an end or without an end.

(114) A depth position of the floating dummy pattern inside the insulating layer **51** can be any as desired, and is adjusted according to an intensity of an electric field to be alleviated.

(115) The number of the floating lines can be any as desired, and is adjusted according to an intensity of an electric field to be alleviated. The floating dummy pattern can also be formed of a plurality of floating lines.

(116) Referring to FIG. 7, the semiconductor device **5** includes a second functional component **60** formed on the first main surface **42** of the semiconductor chip **41** in a component region **62**. The second functional component **60** is formed by using a surface layer portion of the first main surface **42** of the semiconductor chip **41** and/or a region on the first main surface **42** of the semiconductor chip **41**, and is covered by the insulating layer **51** (the lowermost insulating layer **55**). In FIG. 7, the dotted lines shown at the surface layer portion of the first main surface **42** schematically represent the second functional component **60**.

(117) The second functional component **60** is electrically connected to the low potential terminal **11** by a low potential wiring, and is electrically connected to the high potential terminal **12** by a high potential wiring. Except for being laid in a round-about manner to connect the second functional component **60** to inside the insulating layer **51**, the low potential wiring has a same structure as the first low potential wiring **31** (the second low potential wiring **32**). Except for being laid in a round-about manner to connect the second functional component **60** to inside the insulating layer **51**, the high potential wiring has a same structure as the first high potential wiring **33** (the second high potential wiring **34**). Specific details of the low potential wiring and the high potential wiring of the second functional component **60** are omitted herein.

(118) The second functional component **60** can include at least one of a passive component, a semiconductor rectifier and a semiconductor switching component. The second functional component **60** can also include a circuit network including a combination of at least two components selected from a passive component, a semiconductor rectifier and a semiconductor switching component. The circuit network can form part or all of an integrated circuit.

(119) The passive component can include a semiconductor passive component. The passive component can also include any one or two of a resistor and a capacitor. The semiconductor rectifier component can include at least one of a pn junction diode, a PIN diode, a Zener diode, a Schottky diode and a fast recovery diode. The semiconductor switching component can include at least one of a bipolar function transistor (BJT), a metal insulator semiconductor field effect

transistor (MISFET), an insulated gate bipolar junction transistor (IGBT) and a junction field effect transistor (JFET).

(120) Referring to FIG. 5 to FIG. 7, the semiconductor device 5 further includes a sealing conductor 61 buried in the insulating layer 51. The sealing conductor 61 is buried at a distance from the insulating sidewalls 53A to 53D in the insulating layer 51, and divides the insulating layer 51 into the component region 62 and an outer region 63. The sealing conductor 61 suppresses moisture and cracks from entering the component region 62 from the outer region 63.

(121) The component region 62 is a region including the first functional component 45 (the plurality of transformers 21), the second functional component 60, the plurality of low potential terminals 11, the plurality of high potential terminals 12, the first low potential wiring 31, the second low potential wiring 32, the first high potential wiring 33, the second high potential wiring 34 and the dummy pattern 85. The outer region 63 is a region outside the component region 62.

(122) The sealing conductor 61 is electrically disconnected from the component region 62. More specifically, the sealing conductor 61 is electrically disconnected from the first functional component 45 (the plurality of transformers 21), the second functional component 60, the plurality of low potential terminals 11, the plurality of high potential terminals 12, the first low potential wiring 31, the second low potential wiring 32, the first high potential wiring 33, the second high potential wiring 34 and the dummy pattern 85. More specifically, the sealing conductor 61 is fixed at an electrically floating state. The sealing conductor 61 does not form any current path connected to the component region 62.

(123) The sealing conductor 61 is formed to have a strip shape extending along the insulating sidewalls 53A to 53D in the plan view. In this implementation, the sealing conductor 61 is formed to have a quadrilateral loop shape (more specifically, a rectangular loop) in the plan view. Accordingly, the sealing conductor 61 defines the component region 62 having a quadrilateral shape (specifically, a rectangular shape) in the plan view. Moreover, the sealing conductor 61 defines the outer region 63 having a quadrilateral loop shape (specifically, a rectangular loop) surrounding the component region 62 in the plan view.

(124) More specifically, the sealing conductor 61 includes an upper end at the side of the insulating main surface 52, a lower end at the side of the semiconductor chip 41, and a wall portion extending as a wall between the upper end and the lower end. In this implementation, the upper end of the sealing conductor 61 is formed at a distance from the insulating main surface 52 toward the side of the semiconductor chip 41, and is located in the insulating layer 51. In this implementation, the upper end of the sealing conductor 61 is covered by the uppermost insulating layer 56. The upper end of the sealing conductor 61 can also be covered by one or more interlayer insulating layers 57. The upper end of the sealing conductor 61 can also be exposed from the uppermost insulating layer 56. The lower end of the sealing conductor 61 is formed at a distance from the semiconductor chip 41 toward the side of the upper end.

(125) As such, in this implementation, the sealing conductor 61 is buried in the insulating layer 51 to be located on the side of the semiconductor chip 41 with respect to the plurality of low potential terminals 11 and the plurality of high potential terminals 12. Moreover, in the insulating layer 51, the sealing conductor 61 faces the first functional component 45 (the plurality of transformers 21), the first low potential wiring 31, the second low potential wiring 32, the first high potential wiring 33, the second high potential wiring 34 and the dummy pattern 85 in a direction in parallel to the insulating main surface 52. In the insulating layer 51, the sealing conductor 61 can also face a part of the second functional component 60 in a direction in parallel to the insulating main surface 52.

(126) The sealing conductor 61 includes a plurality of sealing plug conductors 64, and one or more (multiple in this implementation) sealing path conductors 65. The number of the sealing path conductors 65 can be any as desired. The uppermost sealing plug conductor 64 of the plurality of sealing plug conductors 64 forms the upper end of the sealing conductor 61. The plurality of sealing path conductors 65 individually form the lower end of the sealing conductor 61. The sealing

plug conductors **64** and the sealing path conductors **65** are preferably formed of a same conductive material as the low potential coil **22**. That is to say, each of the sealing plug conductors **64** and the sealing path conductors includes a barrier layer and a body layer, as the low potential coil **22**.

(127) The plurality of sealing plug connectors **64** are respectively buried in the plurality of interlayer insulating layers **57**, and each forms a quadrilateral loop shape (more specifically, a rectangular loop shape) surrounding the component region **62** in the plan view. The plurality of sealing plug conductors **64** are laminated from the lowermost insulating layer **55** toward the uppermost insulating layer **56** to be connected to one another. The number of the plurality of sealing plug conductors **64** laminated is consistent with the number of the plurality of interlayer insulating layers **57** laminated. As a matter of course, one or more sealing plug conductors **64** passing through the plurality of interlayer insulating layers **57** can also be formed.

(128) If one loop-shaped sealing conductor **61** is formed by an integral of the plurality of sealing plug conductors **64**, it is not necessary for all of the plurality of sealing plug conductors **64** to be formed to have a loop shape. For example, at least one of the plurality of sealing plug conductors **64** can be formed to have a shape with an end. Moreover, at least one of the plurality of sealing plug conductors **64** can be divided to include a plurality of strip portions with ends. However, in view of the risk of moisture and cracks entering the component region **62**, the plurality of sealing plug conductors **64** are preferably formed to be shapes without ends (formed as loops).

(129) The plurality of sealing path conductors **65** are formed in a region between the semiconductor chip **41** and the sealing plug conductors **64** in the lowermost insulating layer **55**. The plurality of sealing path conductors **65** are formed at a distance from the semiconductor chip **41**, and are connected to the sealing plug conductors **64**. The plurality of sealing path conductors **65** have a planar area less than a planar area of the sealing plug conductors **64**. When a single sealing path conductor **65** is formed, the single sealing path conductor **65** can also have a planar area greater than the planar area of the sealing plug conductors **64**.

(130) A width of the sealing conductor **61** can be between about 0.1  $\mu\text{m}$  and about 10  $\mu\text{m}$ . The width of the sealing conductor **61** is preferably between about 1  $\mu\text{m}$  and about 5  $\mu\text{m}$ . The width of the sealing conductor **61** is determined by a width in a direction perpendicular to an extension direction of the sealing conductor **61**.

(131) Referring to FIG. 7 and FIG. 8, the semiconductor device **5** further includes a separation structure **130**, which is located between the semiconductor chip **41** and the sealing conductor **61** and electrically disconnects the sealing conductor **61** and the semiconductor chip **41**. The separation structure **130** preferably includes an insulator. In this implementation, the separation structure **130** includes a field insulating film **131** formed on the first main surface **42** of the semiconductor chip **41**.

(132) The field insulating film **131** includes at least one of an oxide film (a SiO film) and a nitride film (a SiN film). The field insulating film **131** preferably includes a local oxidation of silicon (LOCOS) film, which is an example of an oxide film formed by oxidizing the first main surface **42** of the semiconductor chip **41**. A thickness of the field insulating film **131** can be any as desired, provided that the semiconductor chip **41** and the sealing conductor **61** can be insulated. The thickness of the field insulating film **13** can be between about 0.1  $\mu\text{m}$  and about 5  $\mu\text{m}$ .

(133) The separation structure **130** is formed on the first main surface **42** of the semiconductor chip **41**, and extends as a strip shape along the sealing conductor **61** in the plan view. In this implementation, the separation structure **130** is formed to have a quadrilateral loop shape (more specifically, a rectangular loop) in the plan view. The separation structure **130** includes a connection portion **132** connected to the lower end of the sealing conductor **61** (the sealing path conductor **65**). The connecting portion **132** can also form an anchor portion for which the lower end of the sealing conductor **61** (the sealing via conductor **65**) bites toward the side of the semiconductor chip **41**. As a matter of course, the connecting portion **132** can also be formed as a leveled surface with respect to a main surface of the separation structure **130**.



(134) The separation structure **130** includes an inner end **130A** on the side of the component region **62**, an outer end **130B** on the side of the outer region **63**, and a body **130C** between the inner end **130A** and the outer end **130B**. The inner end **130A** defines a region in which the second functional component **60** is formed (that is, the component region **62**) in the plan view. The inner end **130A** can also be formed as an integral with an insulating film (not shown) formed on the first main surface **42** of the semiconductor chip **41**.

(135) The outer end **130B** is exposed from the chip sidewalls **44A** to **44D** of the semiconductor chip **41**, and is connected to the chip sidewalls **44A** to **44D** of the semiconductor chip **41**. More specifically, the outer end **130B** is formed as a leveled surface relative to the chip sidewalls **44A** to **44D** of the semiconductor chip **41**. The outer end **130B** is formed as a leveled grinding surface between the chip sidewalls **44A** to **44D** of the semiconductor chip **41** and the insulating sidewalls **53A** to **53D** of the insulating layer **51**. In other implementations, the outer end **130B** can also be formed at a distance from the chip sidewalls **44A** to **44D** in the first main surface **42**.

(136) The body **130C** has a flat surface extending nearly in parallel to the first main surface **42** of the semiconductor chip **41**. The body **130C** includes a connection portion **132** connected to the lower end of the sealing conductor **61** (the sealing path conductor **65**). The connecting portion **132** is formed at a distance from the inner end **130A** and the outer end **130B** in a part of the body **130C**. The separation structure **130** can be implemented in various forms instead of the field insulating film **131**.

(137) Referring to **7**, the semiconductor device **5** further includes an inorganic insulating layer **140** formed on the first main surface **52** of the semiconductor chip **51** to cover the sealing conductor **61**. The organic insulating layer **140** can also be referred to as a passivation layer. The inorganic insulating layer **140** protects the insulating layer **51** and the semiconductor chip **41** on the insulating main surface **52**.

(138) In this implementation, the inorganic insulating layer **140** has a laminated structure including a first inorganic insulating layer **141** and a second inorganic insulating layer **142**. The first inorganic insulating layer **141** can include SiO. The first inorganic insulating layer **141** is preferably undoped silicate glass (USG) which is SiO without any impurity. A thickness of the first inorganic insulating layer **141** can be between about 50  $\mu\text{m}$  and about 5000  $\mu\text{m}$ . The second inorganic insulating layer **142** can include SiN. A thickness of the second inorganic insulating layer **142** can be between about 500  $\mu\text{m}$  and about 5000  $\mu\text{m}$ . By increasing a total thickness of the inorganic insulating layer **140**, the insulation withstand voltage on the high potential coil **23** can be increased.

(139) When the first inorganic insulating layer **141** includes USG and the second inorganic insulating layer **142** includes SiN, an insulation breakdown voltage (V/cm) of the USG is greater than an insulation breakdown voltage (V/cm) of the SiN. Thus, when the inorganic insulating layer **140** is designed to be thick, the first inorganic insulating layer **141** is preferably formed to be thicker than the second inorganic insulating layer **142**.

(140) The first inorganic insulating layer **141** can include at least one of boron doped phosphor silicate glass and phosphorous silicate glass (PSG) as an example of SiO. However, in this case, since the SiO includes an impurity (boron or phosphor), the first inorganic insulating layer **141** containing USG is preferably formed in order to increase the insulation withstand voltage on the high potential coil **23**. As a matter of course, the inorganic insulating layer **140** can also be a single-layer structure which includes any one between the first inorganic insulating layer **141** and the second inorganic insulating layer **142**.

(141) The inorganic insulating layer **140** covers all of the sealing conductor **61**, and includes a plurality of low potential pad openings **143** and a plurality of high potential pad openings **144** formed in a region outside the sealing conductor **61**. The plurality of low potential pad openings **143** respectively expose the plurality of low potential terminals **11**. The plurality of high potential pad openings **143** respectively expose the plurality of high potential terminals **12**. The inorganic

insulation layer **140** can include an overlapping portion which climbs up a peripheral portion of the low potential terminals **11**. The inorganic insulation layer **140** can also include an overlapping portion which climbs up a peripheral portion of the high potential terminals **12**.

(142) The semiconductor device **5** further includes an organic insulating layer **145** formed on inorganic insulation layer **140**. The organic insulation layer **145** can include a photosensitive resin. The organic insulating layer **145** can also include at least one of polyimide, polyamide and polybenzoxazole. In this implementation, the organic insulating layer **145** includes polyamide. A thickness of the organic insulating layer **145** can be between about 1  $\mu\text{m}$  and about 50  $\mu\text{m}$ .

(143) The thickness of the organic insulating layer **145** is preferably greater than the total thickness of the inorganic insulating layer **140**. Moreover, a total thickness of the inorganic insulation layer **140** and the organic insulation layer **145** is preferably greater than the distance **D2** between the low potential coil **22** and the high potential coil **23**. In this case, the total thickness of the inorganic insulating layer **140** is preferably between about 2  $\mu\text{m}$  and about 10  $\mu\text{m}$ . Moreover, the thickness of the organic insulating layer **145** is preferably between about 5  $\mu\text{m}$  and about 50  $\mu\text{m}$ . According to the structures above, thickening of the inorganic insulating layer **140** and the organic insulating layer **145** can be suppressed, and at the same time the insulation withstand voltage on the high potential coil **23** can be appropriately increased by the laminated films of the inorganic insulating layer **140** and the organic insulating film **145**.

(144) The organic insulating layer **145** includes a first portion **146** covering a region on the low potential side and a second portion **147** covering a region on the high potential side. The first portion **146** is separated by the inorganic insulating layer **140** to cover the sealing conductor **61**. The first portion **146** includes a plurality of low potential terminal openings **148** exposing the plurality of low potential terminals **11** (the low potential pad openings **143**) in a region outside the sealing conductor **61**. The first portion **146** can also include an overlapping portion which climbs up a periphery (an overlapping portion) of the low potential pad openings **143**.

(145) The second portion **147** is formed at a distance from the first portion **146**, and exposes the inorganic insulating layer **140** from amidst the first portion **146**. The second portion **147** includes a plurality of high potential terminal openings **149** exposing the plurality of high potential terminals **12** (the high potential pad openings **144**)<sup>1</sup>. The second portion **147** can also include an overlapping portion which climbs up a periphery (an overlapping portion) of the high potential pad openings **144**.

(146) The second portion **147** universally covers the transformers **21A** to **21D** and the dummy pattern **85**. More specifically, the second portion **147** universally covers the plurality of high potential coils **23**, the plurality of high potential terminals **12**, the first high potential dummy pattern **87** (including a first pattern **93**, a second pattern **94** and a third pattern **95**), the second high potential dummy pattern **88** and the floating dummy pattern **121**.

(147) The embodiments of the present disclosure can also be implemented in form of other embodiments. In the embodiment, an example in which the first functional component **45** and the second functional component **60** are formed is described. However, an implementation without the first functional component **45** but including only the second functional component **60** can also be practiced. In this case, the dummy pattern **85** can be removed. According to the structure above, regarding the second functional component **60**, the same effects as those of the first embodiment can be achieved.

(148) That is to say, when a voltage is applied to the second functional component **60** through the low potential terminal **11** and the high potential terminal **12**, an unexpected conduction between the high potential terminal **12** and the sealing conductor **61** can be suppressed. Moreover, when a voltage is applied to the second functional component **60** through the low potential terminal **11** and the high potential terminal **12**, an unexpected conduction between the low potential terminal **11** and the sealing conductor **61** can be suppressed.

(149) Moreover, in the embodiments, an example in which the second functional component **60** is

formed is described. However, the second functional component **60** is not necessary and can also be omitted.

(150) In the embodiments, an example in which the dummy pattern **85** is formed is described. However, the dummy pattern **85** is not necessary and can also be omitted.

(151) Moreover, in the embodiments, an example in which the first functional component **45** is configured as a multi-channel type including the plurality of transformers **21** is described. However, the first functional component **45** configured as a single-channel type including a single transformer **21A** can also be used.

(152) <Arrangement of Transformers>

(153) FIG. **9** shows a plan view (a top view) of an example of an arrangement of transformers in a two-channel transformer chip **300** (equivalent to the semiconductor device **5** above). The transformer chip **300** of this drawing includes a first transformer **301**, a second transformer **302**, a third transformer **303**, a fourth transformer **304**, a first protection ring **305**, a second protection ring **306**, pads **a1** to **a8**, pads **b1** to **b8**, pads **c1** to **c4** and pads **d1** to **d4**.

(154) In the transformer chip **300**, the pads **a1** and **b1** are connected to one end of a secondary-side coil **L1s** forming the first transformer **301**, and the pads **c1** and **d1** are connected to the other end of the secondary-side coil **L1s**. The pads **a2** and **b2** are connected to one end of a secondary-side coil **L2s** forming the second transformer **302**, and the pads **c1** and **d1** are connected to the other end of the secondary-side coil **L2s**.

(155) Moreover, the pads **a3** and **b3** are connected to one end of a secondary-side coil **L3s** forming the third transformer **303**, and the pads **c2** and **d2** are connected to the other end of the secondary-side coil **L3s**. The pads **a4** and **b4** are connected to one end of a secondary-side coil **L4s** forming the fourth transformer **304**, and the pads **c2** and **d2** are connected to the other end of the secondary-side coil **L4s**.

(156) Moreover, a primary-side coil forming the first transformer **301**, a primary-side coil forming the second transformer **302**, a primary-side coil forming the third transformer **303** and a primary-side coil forming the fourth transformer **304** are not shown in this drawing. However, the primary-side coils respectively have the same structures as the secondary-side coils **L1s** to **L4s**, and are arranged directly below the respective secondary-side coils **L1s** to **L4s** to face these secondary-side coils **L1s** to **L4s**, respectively.

(157) That is to say, the pads **a5** and **b5** are connected to one end of the primary-side coil forming the first transformer **301**, and the pads **c3** and **d3** are connected to the other end of the primary-side coil. Moreover, the pads **a6** and **b6** are connected to one end of the primary-side coil forming the second transformer **302**, and the pads **c3** and **d3** are connected to the other end of the primary-side coil.

(158) Moreover, the pads **a7** and **b7** are connected to one end of the primary-side coil forming the third transformer **303**, and the pads **c4** and **d4** are connected to the other end of the primary-side coil. Moreover, the pads **a8** and **b8** are connected to one end of the primary-side coil forming the fourth transformer **304**, and the pads **c4** and **d4** are connected to the other end of the primary-side coil.

(159) However, the pads **a5** to **a8**, the pads **b5** to **b8**, the pads **c3** and **c4** and the pads **d3** and **d4** are led out to a surface from the inside of the transformer chip **300** via a path not shown in the drawing.

(160) The pads **a1** to **a8** of the plurality of pads are equivalent to first current supply pads, and the pads **b1** to **b8** are equivalent to first voltage measurement pads. Moreover, the pads **c1** to **c4** are equivalent to second current supply pads, and the pads **d1** to **d4** are equivalent to second voltage measurement pads.

(161) Hence, with respect to the transformer chip **300** of this configuration example, a series resistance component of each coil can be accurately measured during an inspection of defective products. Thus, in addition to avoiding production of defective products with disconnections in each coil, production of defective products with abnormal resistance values of each coil (for

example, short-circuit among coils) can be appropriately avoided, further preventing in advance defective products from being available on the market.

(162) In addition, for the transformer chip **300** having passed the inspection of defective products, the plurality of pads are used as a connection mechanism between a primary-side chip and a secondary-side chip (for example, the controller chip **210** and the driver **220** described above).

(163) More specifically, the pads **a1** and **b1**, the pads **a2** and **b2**, the pads **a3** and **b3**, and the pads **a4** and **b4** may be respectively connected to signal input terminals or signal output terminals of the secondary-side chip. Moreover, each of the pads **c1**, **d1** and the pads **c2**, **d2** may be connected to a common voltage application end (GND2) of the secondary-side chip.

(164) Moreover, the pads **a5** and **b5**, the pads **a6** and **b6**, the pads **a7** and **b7**, and the pads **a8** and **b8** may be respectively connected to signal input terminals or signal output terminals of the primary-side chip. Moreover, each of the pads **c3**, **d3** and the pads **c4**, **d4** may be connected to a common voltage application end (GND1) of the primary-side chip.

(165) Herein, as shown in FIG. **9**, the first transformer **301** to the fourth transformer **304** perform coupling according to each of the signal transmission directions and are thus arranged. According to the illustration of this drawing, for example, the first transformer **301** and the second transformer **302** which transmit signals from the primary-side chip to the secondary-side chip become a first pair via the first protection ring **305**. Moreover, for example, the third transformer **303** and the fourth transformer **304** which transmit signals from the secondary-side chip to the first-side chip become a second pair via the second protection ring **306**.

(166) The reason for the coupling is that, when primary-side coils and secondary-side coils forming the first transformer **301** to the fourth transformer **304** are laminated in a manner of being stacked in a vertical direction of the substrate of the transformer chip **300**, a withstand voltage between the primary-side coils and the secondary-side coils is ensured. However, the first protection ring **305** and the second protection ring **306** are not necessary constituting elements.

(167) In addition, the first protection ring **305** and the second protection ring **306** are connected to a low impedance wiring such as a ground terminal via the pads **e1** and **e2**, respectively.

(168) Moreover, in the transformer chip **300**, the pads **c1** and **d1** are shared between the secondary-side coil **L1s** and the secondary-side coil **L2s**. Moreover, the pads **c2** and **d2** are shared between the secondary-side coil **L3s** and the secondary-side coil **L4s**. Moreover, the pads **c3** and **d3** are shared between the primary-side coil **L1p** and the primary-side coil **L2p**. Moreover, the pads **c4** and **d4** are shared between the corresponding primary-side coils. With the setting of the configuration above, the number of pads can be reduced to implement miniaturization of the transformer chip **300**.

(169) Moreover, as shown in FIG. **9**, it is desired that each of the primary-side coils and the secondary-side coils of the first transformer **301** to the fourth transformer **304** is wound to have a rectangular shape (or a track with rounded corners) in a plan view of the transformer chip **300**. With the setting of the configuration above, an area of an overlapping portion of the primary-side coils and the secondary-side coils is increased, hence improving transmission efficiency of the transformers.

(170) It should be noted that the arrangement of the transformers in this drawing is merely an example, and the number, shape and configuration of the coils as well as the arrangement of the pads can be any as desired. Moreover, the chip structure and the arrangement of the transformers described so far are applicable to an entire semiconductor device in which coils are integrated on the semiconductor chip.

(171) <Signal Transmission Device (Brief Configuration)>

(172) FIG. **10** shows a diagram of a general brief configuration of a signal transmission device (a so-called integrated circuit (IC)). The signal transmission device **400** of this configuration example includes a transmitting circuit **410**, an insulated transmission path **420** and a receiving circuit **430**.

(173) The transmitting circuit **410** receives an input pulse signal **IN** and outputs a transmission pulse signal **S31**.

(174) The insulated transmission path **420** insulates the transmitting circuit **410** and the receiving circuit **430**, and transmits the transmission pulse signal **S31** output from the transmitting circuit **410** as a reception pulse signal **S32** to the receiving circuit **430**. The insulated transmission path **420** includes an insulating transmission element such as a coil (for example, a transformer), a capacitor or an optical component (for example, an optical coupler).

(175) The receiving circuit **430** receives the reception pulse signal **S32** output from the insulated transmission path **420** and outputs an output pulse signal **OUT**. In addition, the output pulse signal **OUT** can also be a drive pulse signal for driving a motor to rotate.

(176) However, a component for insulated communications is required to have higher reliability. For example, the signal transmission device **400** mounted on an industrial machine or vehicle is required to continue signal transmission in the event of a malfunction of the insulated transmission path **420** so that the system can operate safely.

(177) <Signal Transmission Device (First Comparison Example)>

(178) FIG. **11** shows a diagram of a first comparison example (equivalent to a malfunction countermeasure compared with an embodiment described below) of the signal transmission device **400**.

(179) By using the previous brief configuration (FIG. **10**) as a basis, the signal transmission device **400** of this comparison example includes a plurality of insulated transmission paths (insulated transmission paths **421** and **422** in this drawing) as the insulated transmission path **420**. Moreover, the insulated transmission path **421** is a main transmission path, while the insulated transmission path **422** is a reserved transmission path. That is to say, in the signal transmission device **400** of the first comparison example, the insulated transmission path **420** is multiplexed (duplexed in this drawing) in a redundant manner.

(180) For example, when the insulated transmission path **421** does not contain any malfunction, as shown by the dotted arrow in this drawing, signal transmission is performed via the insulated transmission path **421**. That is to say, the insulated transmission path **421** transmits the transmission pulse signal **S31** output from the transmitting circuit **410** as a reception pulse signal **S32** to the receiving circuit **430**. As a result, the input pulse signal **IN** (equivalent to a data string **A1A2A3A4**, wherein each of **A1** to **A4** is a 1-bit (H/L) data) is used as the output pulse signal **OUT** (equivalent to an output data string **A1A2A3A4**) and transmitted.

(181) FIG. **12** shows a diagram of behaviors during a malfunction of the insulated transmission path **421** in the signal transmission device **400** of the first comparison example.

(182) In the event of a malfunction of the insulated transmission path **421** due to component degradation or a malfunction from an external component, as shown by the dotted line in this drawing, signal transmission is performed via the reserved insulating transmission path **422**. That is to say, the insulated transmission path **422** transmits the transmission pulse signal **S31** output from the transmitting circuit **410** as a reception pulse signal **S32** to the receiving circuit **430**. As a result, the input pulse signal **IN** (equivalent to the data string **A1A2A3A4**) is used as the output pulse signal **OUT** (equivalent to the output data string **A1A2A3A4**) and transmitted.

(183) With regard to the signal transmission device **400** of the first comparison example, even if the main insulated transmission path **421** malfunctions, signal transmission can be continued by using the reserved insulated transmission path **422**. However, the reserved insulated transmission path **422** is still needed even when there is no malfunction in the main insulated transmission path **421**. Thus, an increase in costs of the signal transmission device **400** is resulted.

(184) <Signal Transmission Device (Second Comparison Example)>

(185) FIG. **13** shows a diagram of a second comparison example (equivalent to two-channel configuration example compared with an embodiment described below) of the signal transmission device **400**.

(186) Similar to the first comparison example (FIG. **11**), the signal transmission device **400** of the second comparison example includes a plurality of insulated transmission paths (insulated

transmission paths **421** and **422** in this drawing) as the insulated transmission path **420**. However, the insulated transmission path **422** is not a reserve of the insulated transmission path **421**. The signal transmission device **400** uses both of the insulated transmission paths **421** and **422** to implement signal transmission in parallel in a same direction.

(187) The transmitting circuit **410** receives an input pulse signal INA (equivalent to the data string **A1A2A3A4**) of a first channel, and outputs a transmission pulse signal **S31A** (equivalent to a transmission data string **A1A2A3A4**) of the first channel.

(188) Moreover, the transmitting circuit **410** receives an input pulse signal INB (equivalent to a data string **B1B2B3B4**) of a second channel, and outputs a transmission pulse signal **S31B** (equivalent to a transmission data string **B1B2B3B4**) of the second channel.

(189) The insulated transmission path **421** insulates between the transmitting circuit **410** and the receiving circuit **430**, and transmits the transmission pulse signal **S31A** of the first channel output from the transmitting circuit **410** as a reception pulse signal **S32A** of the first channel to the receiving circuit **430**.

(190) Moreover, the insulated transmission path **422** insulates between the transmitting circuit **410** and the receiving circuit **430**, and transmits the transmission pulse signal **S31B** of the second channel output from the transmitting circuit **410** as a reception pulse signal **S32B** of the second channel to the receiving circuit **430**.

(191) The receiving circuit **430** receives the reception pulse signal **S32A** of the first channel output from the insulated transmission path **421** and outputs an output pulse signal OUTA (equivalent to the output data string **A1A2A3A4**) of the first channel.

(192) Moreover, the receiving circuit **430** receives the reception pulse signal **S32B** of the second channel output from the insulated transmission path **422** and outputs an output pulse signal OUTB (equivalent to the output data string **B1B2B3B4**) of the second channel.

(193) In the signal transmission device **400** of the second comparison example, if at least one between the insulated transmission paths **421** and **422** malfunctions, signal transmission cannot be correctly performed. Moreover, if multiplexing as the insulated transmission path **420** of the first comparison example (FIG. 1) is implemented, continuity of signal transmission can be ensured. However, as described above, the issue lies in that an increase in costs of the signal transmission device **400** is resulted.

(194) <Signal Transmission Device (Implementation)>

(195) FIG. 14 shows a diagram of the signal transmission device **400** in a novel embodiment. The signal transmission device **400** of this embodiment uses the second comparison example (FIG. 13) as a basis, and modifies respective internal signal processing of the transmitting circuit **410** and the receiving circuit **430**.

(196) The transmitting circuit **410** generates a transmission pulse signal **S31X** (equivalent to a mixed data string **A1B2A3B4**) according to an input pulse signal INA (equivalent to a data string **A1A2A3A4**) and an input pulse signal INB (equivalent to a data string **B1B2B3B4**).

(197) Moreover, the transmitting circuit **410** generates a transmission pulse signal **S31Y** (equivalent to a mixed data string **B1A2B3A4**) according to the input pulse signal INA (equivalent to the data string **A1A2A3A4**) and the input pulse signal INB (equivalent to the data string **B1B2B3B4**).

(198) As illustrated according to this drawing, for the data **A1** to **A4** included in the input pulse signal INA (equivalent to the data string **A1A2A3A4**) and the data **B1** to **B4** included in the input pulse signal INB (equivalent to the mixed data string **B1A2B3A4**), sequentially switches from the beginning and outputs each data according to every bit to one of the insulated transmission paths **421** and **422**.

(199) That is to say, the transmitting circuit **410** alternately switches an output destination of each of the data **A1** to **A4** and the data **B1** to **B4**, while generating the transmission pulse signals **S31X** and **S31Y**. Thus, the transmission pulse signal **S31X** of the first channel becomes the mixed data

string **A1B2A3B4** including the data **A1** and **A3** of the first channel and the data **B2** and **B4** of the second channel. Moreover, the transmission pulse signal **S31Y** of the second channel becomes the mixed data string **B1A2B3A4** including the data **A2** and **A4** of the first channel and the data **B1** and **B3** of the second channel.

(200) The insulated transmission path **421** insulates between the transmitting circuit **410** and the receiving circuit **430**, and transmits the transmission pulse signal **S31X** of the first channel output from the transmitting circuit **410** as a reception pulse signal **S32X** of the first channel to the receiving circuit **430**. That is to say, the mixed data string **B1A2B3A4** is transmitted to the insulating transmission path **421**.

(201) Moreover, the insulated transmission path **422** insulates between the transmitting circuit **410** and the receiving circuit **430**, and transmits the transmission pulse signal **S31Y** of the second channel output from the transmitting circuit **410** as a reception pulse signal **S32B** of the second channel to the receiving circuit **430**. That is to say, the mixed data string **B1A2B3A4** is transmitted to the insulating transmission path **422**.

(202) The receiving circuit **430** restores a first output signal **OUTA** (equivalent to the data string **A1A2A3A4**) equivalent to the input pulse signal **INA** of the first channel according to the reception pulse signal **S32X** (equivalent to the mixed data string **A1B2A3B4**) of the first channel and the reception pulse signal **S32Y** (equivalent to the mixed data string **B1A2B3A4**) of the second channel.

(203) The receiving circuit **430** restores a second output signal **OUTB** (equivalent to the data string **B1B2B3B4**) equivalent to the input pulse signal **INB** of the second channel according to the reception pulse signal **S32X** (equivalent to the mixed data string **A1B2A3B4**) of the first channel and the reception pulse signal **S32Y** (equivalent to the mixed data string **B1A2B3A4**) of the second channel.

(204) For example, a case where neither of the insulated transmission paths **421** and **422** malfunctions is taken into consideration. In this case, as shown in this drawing, the mixed data string **A1B2A3B4** is transmitted in an insulated manner via the insulated transmission path **421**. Moreover, the mixed data string **B1A2B3A4** is transmitted in an insulated manner via the insulated transmission path **422**. Thus, in the receiving circuit **430**, the output pulse signals **OUTA** and **OUTB** without any data loss are restored. As a result, the input pulse signal **IN** (equivalent to the data string **A1A2A3A4**) is used as the output pulse signal **OUT** (equivalent to an output data string **A1A2A3A4**) and transmitted correctly.

(205) FIG. 15 shows a diagram of behaviors during a malfunction of the insulated transmission path **422** in the signal transmission device **400** of this embodiment.

(206) In the event of a malfunction of the insulated transmission path **422** due to component degradation or a malfunction from an external component, the mixed data string **B1A2B3A4** that should be transmitted via the insulated transmission path **422** is lost, while the mixed data string **A1B2A3B4** is transmitted to the receiving circuit **430** via the malfunction-free insulated transmission path **421**. Thus, in the receiving circuit **430**, although part of the data is lost, the output pulse signals **OUTA** and **OUTB** can still be restored.

(207) According to the illustration of this drawing, the receiving circuit **430** receives the reception pulse signal **S32X** of the first channel output from the insulated transmission path **421** and outputs an output pulse signal **OUTA** (equivalent to the output data string **A1(A1)A3(A3)**) of the first channel.

(208) Moreover, the receiving circuit **430** receives the reception pulse signal **S32X** of the first channel output from the insulated transmission path **421** and outputs an output pulse signal **OUTB** (equivalent to the output data string **B2(B2)B4(B4)**) of the second channel.

(209) As such, with regard to the signal transmission device **400** of this embodiment, even if one between the insulated transmission paths **421** and **422** malfunctions, only a state in which the communication speed is reduced without incurring a complete halt in functions is resulted, and so

signal transmission can be continued. Thus, even if the insulated transmission path **420** malfunctions, signal transmission can be continued and the system (an industrial machine or a vehicle) is able to continue operating safely.

(210) Moreover, different from the first comparison example (FIG. **11**), the signal transmission device **400** of this embodiment does not need a reserve insulating transmission path. Thus, an increase in costs of the signal transmission device **400** is not resulted.

(211) <Observations Related to Switching Timing of Data>

(212) In order to implement internal signal processing, as described above, the output destination of each of the data **A1** to **A4** included in the input pulse signal INA (equivalent to the data string **A1A2A3A4**) and the data **B1** to **B4** included in the input pulse signal INB (equivalent to the data string **B1B2B3B4**) needs to be alternately switched according to a predetermined switching timing.

(213) In addition, an entity alternately switching the output destination of data **A1** to **A4** and an entity alternately switching the output destination of the data **B1** to **B4** are both the transmitting circuit **410**. Thus, the switching timing in the transmitting circuit **410** can be set as desired.

(214) On the other hand, the receiving circuit **430** operates asynchronously from the transmitting circuit **410**. Thus, for the receiving circuit **430** to restore the output pulse signals OUTA and OUTB, the receiving circuit **430** needs to be notified of the switching timing from the transmitting circuit **410**.

(215) As a simplest approach, another additional insulated transmission path solely for the notification of the switching timing is added. However, in the approach above, similar to the first comparison example (FIG. **11**), an increase in costs of the signal transmission device **400** can be resulted. Moreover, in the event of a malfunction of the insulated transmission path for the notification of the switching timing, restoration of both of the output pulse signals OUTA and OUTB cannot be achieved and thus signal transmission cannot be continued.

(216) Thus, in view of the observations above, the transmitting circuit **410** adds a switching timing flag STF to each data while generating the transmission pulse signal **S31X** (equivalent to the data string **A1B2A3B4**). Moreover, the transmitting circuit **410** adds the switching timing flag STF to each data while generating the transmission pulse signal **S31Y** (equivalent to the data string **B1A2B3A4**).

(217) The receiving circuit **430** restores the output pulse signals OUTA and OUTB according to the switching timing flag STF added for each data.

(218) <Switching Timing Flag (First Example)>

(219) FIG. **16** shows a diagram of a first example of the switching timing flag STF. In this drawing, the switching timing flag STF is added to a header of a data **D0**. In addition, the switching timing flag STF of the first example includes flag pulses **P0** and **P1** having different logic levels. For example, the flag pulse **P0** can be at a high level and the flag pulse **P1** can be at a low level.

(220) As shown by the upper part of this drawing, when the data **D0** is at a low level, transmission is performed in a continuous manner of the flag pulse **P0** at a high level, the flag pulse **P1** at a low level and the data **D0** at a low level. On the other hand, as shown by the lower part of this drawing, when the data **D0** is at a high level, transmission is performed in a continuous manner of the flag pulse **P0** at a high level, the flag pulse **P1** at a low level and the data **D0** at a high level.

(221) That is to say, the switching timing flag STF does not depend on the logic level of the data **D0**, and always includes a pulse edge from a high level to a low level (a falling edge). Thus, the receiving circuit **430** can learn the timing for switching the output destination of the data by detecting the pulse edge.

(222) <Receiving Circuit>

(223) FIG. **17** shows a diagram of a configuration example of the receiving circuit **430**. The receiving circuit **430** of this configuration example includes an edge detection unit **431**, a delay unit **432**, a clock separation unit **433**, a data import unit **434** and an output timing adjustment unit **435**.

(224) The edge detection circuit **431** receives a reception pulse signal Sa transmitted from the



transmitting circuit **410** via the insulated transmission path **421** (or **422**). Moreover, the reception pulse signal Sa is equivalent to the reception pulse signal S32X (or S32Y) described above. The edge detection unit **431** detects the pulse edge of the switching timing flag STF included in the pulse signal Sa, and generates an edge detection signal Sd. Moreover, the edge detection unit **431** internally generates an inverted reception signal Sb and a delayed reception pulse signal Sc (with associated details to be described shortly) while generating the edge detection signal Sd.

(225) The delay unit **432** generates a delayed edge detection signal Se by applying a predetermined delay to the edge detection signal Sd.

(226) The clock separation unit **433** separates the delayed edge detection signal Se into a plurality of clock signals Sg and Sh. Moreover, the clock separation unit **433** internally generates a clock separation signal Sf (with associated details to be described shortly) while generating the clock signals Sg and Sh.

(227) The data import unit **434** imports data included in the reception pulse signal Sa (equivalent to a mixed data string) into either one of separated pulse signals Si and Sj (equivalent to separated data strings) in synchronization with the clock signals Sg and Sh.

(228) The output timing adjustment unit **435** adjusts each of the output timings of the separated pulse signals Si and Sj (equivalent to separated data strings) in synchronization with the clock signals Sg and Sh to generate the output pulse signals OUTA and OUTB (equivalent to output data strings).

(229) FIG. **18** shows a diagram of an example of a signal transmission operation in the receiving circuit **430**. In this drawing, the input pulse signals INA and INB, the reception pulse signal Sa, the inverted reception pulse signal Sb, the delayed reception pulse signal Sc, the edge detection signal Sd, the delayed edge detection signal Se, the clock separation signal Sf, the clock signals Sg and Sh, the separated pulse signals Si and Sj, and the output pulse signals OUTA and OUTB are sequentially depicted from top to bottom.

(230) Moreover, in this drawing, a case where the output pulse signals OUTA and OUTB are restored according to the data string A1B2A3B4 by assuming that the mixed data string B1A2B3A4 is damaged or lost due to a malfunction of the insulated transmission path **422** is depicted.

(231) The input pulse signal INA includes the input data string A1A2A3A4. The data A1 and A2 are both at a low level. The data A3 and A4 are both at a high level.

(232) The input pulse signal INB includes the input data string B1B2B3B4. The data B1 and B2 are both at a high level. The data B3 and B4 are both at a low level.

(233) The reception pulse signal Sa includes the mixed data string A1B2A3B4 sequentially mixed with the data A1 and A3 of the input pulse signal INA and the data B2 and B4 of the input pulse signal INB. That is to say, the reception pulse signal Sa is equivalent to the reception pulse signal S32X (FIG. **14** and FIG. **15**) of the first channel. Moreover, the header of each of the data A1, B2, A3 and B4 is added with the flag pulses P0 and P1 (equivalent to the timing flags STF).

(234) The inverted reception pulse signal Sb is a signal obtained by inverting the logic level of the reception pulse signal Sa. Thus, the inverted reception pulse signal Sb is at a low level when the reception pulse signal Sa is at a high level, and is at a high level when the reception pulse signal Sa is at a low level.

(235) The delayed reception pulse signal Sc is a signal obtained by applying a predetermined delay d1 to the reception pulse signal Sa. Thus, after the reception pulse signal Sa rises from a low level to a high level, the delayed reception pulse signal Sc rises from a low level to a high level at a timing after the delay d1 has elapsed. Moreover, after the reception pulse signal Sa drops from a high level to a low level, the delayed reception pulse signal Sc drops from a high level to a low level at a timing after the delay d1 has elapsed.

(236) The edge detection signal Sd is a logical product signal of the inverted reception pulse signal Sb and the delayed reception pulse signal Sc. Thus, the edge detection signal Sd is at a low level

when at least one between the inverted reception pulse signal Sb and the delayed reception pulse signal Sc is at a low level. Moreover, the edge detection signal Sd is at a high level when both of the inverted reception pulse signal Sb and the delayed reception pulse signal Sc are at a high level. As a result, the edge detection signal Sd becomes a pulse signal which rises to a high level at switching timings of the flag pulses P0 and P1 (equivalent to a generation timing of a falling edge). (237) The delayed edge detection signal Se is a signal obtained by applying a predetermined delay d2 to the edge detection signal Sd. Thus, after the edge detection signal Sd rises from a low level to a high level, the delayed edge detection signal Se rises from a low level to a high level at a timing after the delay d2 has elapsed. Moreover, after the edge detection signal Sd drops from a high level to a low level, the delayed edge detection signal Se drops from a high level to a low level at a timing after the delay d2 has elapsed.

(238) The clock separation signal Sf is a signal of which a logic level is inverted and triggered by a pulse edge of the delayed edge detection signal Se (in this drawing, a falling edge from a high level to a low level). Thus, the clock separation signal Sf switches to a high level when the pulse edge of the delayed edge detection signal Se arrives while the clock separation signal Sf is at a low level. Moreover, the clock separation signal Sf switches to a low level when the pulse edge of the delayed edge detection signal Se arrives while the clock separation signal Sf is at a high level.

(239) The clock signal Sg is a logical product signal of a logical inversion signal of the clock separation signal Sf and the delayed edge detection signal Se. Thus, the delayed edge detection signal Se is transmitted as the clock signal Sg when the clock separation signal Sf is at a low level, while the clock signal Sg is fixed at a low level when the clock separation signal Sf is at a high level. That is to say, among the plurality of pulses generated by the delayed edge detection signal Se, pulses generated during a low level period of the clock separation signal Sf are distributed as pulses of the clock signal Sg.

(240) The clock signal Sh is a logical product signal of a logical product signal of the clock separation signal Sf and the delayed edge detection signal Se. Thus, the delayed edge detection signal Se is transmitted as the clock signal Sh when the clock separation signal Sf is at a high level, while the clock signal Sh is fixed at a low level when the clock separation signal Sf is at a low level. That is to say, among the plurality of pulses generated by the delayed edge detection signal Se, pulses generated during a high level period of the clock separation signal Sf are distributed as pulses of the clock signal Sh.

(241) When the separation clock signal Si is generated, by using a pulse of the clock signal Sg is a trigger, the first data A1 and the third data A3 are sequentially latched from the reception pulse signal Sa (equivalent to the mixed data string A1B2A3B4) of the first channel.

(242) When the separation clock signal Sj is generated, by using a pulse of the clock signal Sh is a trigger, the second data B2 and the fourth data B4 are sequentially latched from the reception pulse signal Sa (equivalent to the mixed data string A1B2A3B4) of the first channel.

(243) The output pulse signals OUTA and OUTB are adjusted signals so that the respective output timings of the separated pulse signals Si and Sj are consistent. As a result, in the receiving circuit 430, although part of the data is lost, the output pulse signals OUTA and OUTB can still be restored. According to the illustration of this drawing, as the output pulse signal OUTA of the first channel, the data string A1(A1)A3(A3)) is output. Moreover, as the output pulse signal OUTB of the second channel, the data string B2(B2)B4(B4) is output.

(244) <Switching Timing Flag (Second Example)>

(245) FIG. 19 shows a diagram of a second example of the switching timing flag STF. In the switching timing flag STF of the second example, different from the first example (FIG. 16), the flag pulse P0 is at a low level and the pulse flag P1 is at a high level. As such, the switching timing flag STF can include a pulse edge from a low level to a high level (a rising edge).

(246) <Variation Example of Flag Adding Operation>

(247) FIG. 20 shows a diagram of a variation example of a flag adding operation. In the flag adding

operation above (FIG. 16 and FIG. 19), a two-bit switching timing flag STF is added to the one-bit data D0. On the other hand, in this variation example, a two-bit switching timing flag STF is added to the two-bit data D0 and D1.

(248) For example, as shown by the upper part of this drawing, when the data D0 and D1 are both at a low level, transmission is performed in a continuous manner of the flag pulse P0 at a high level, the flag pulse P1 at a low level, the data D0 at a low level and the data D1 at a low level.

(249) Moreover, as shown by the second part of this drawing, when the data D0 is at a low level and the data D1 is at a high level, transmission is performed in a continuous manner of the flag pulse P0 at a high level, the flag pulse P1 at a low level, the data D0 at a low level and the data D1 at a high level.

(250) Moreover, as shown by the third part of this drawing, when the data D0 is at a high level and the data D1 is at a low level, transmission is performed in a continuous manner of the flag pulse P0 at a high level, the flag pulse P1 at a low level, the data D0 at a high level and the data D1 at a low level.

(251) Moreover, as shown by the lower part of this drawing, when the data D0 and D1 are both at a high level, transmission is performed in a continuous manner of the flag pulse P0 at a high level, the flag pulse P1 at a low level, the data D0 at a high level and the data D1 at a high level.

(252) As such, the data added with the switching timing flag STF can also be in multiple bits. This variation example can prevent degradation in transmission efficiency caused by the switching timing flag STF added.

(253) For example, as in the flag adding operation above (FIG. 16 and FIG. 19), a case where the two-bit switching timing flag STF added to the one-bit data D0 is taken into consideration. In this case, if the pulse cycle (equivalent to a pulse width) of each of the data 0 and the flag pulses P0 and P1 is the same, compared to a case where no switching timing flag STF is added, the transmission efficiency is reduced to  $\frac{1}{3}$ . On the other hand, with regard to this variation example, the transmission efficiency is only reduced to  $\frac{1}{2}$ . Thus, relatively speaking, the transmission efficiency is improved.

(254) <Switching Timing Flag (Third Example)>

(255) FIG. 21 shows a diagram of a third example of the switching timing flag STF. As shown by the upper part of this drawing, in the first example (FIG. 16) and the second example (FIG. 19) described above, a case where the pulse cycle (equivalent to a pulse width) of each of the data 0 and the flag pulses P0 and P1 is the same is depicted.

(256) On the other hand, as shown by the lower part of this drawing, the flat pulses P0 and P1 can also have cycles shorter than that of the data D0. With such switching timing flag STF, degradation in transmission efficiency caused by the switching timing flag STF added can be prevented.

(257) For example, in a case where the pulse cycle (equivalent to a pulse width) of each of the data 0 and the flag pulses P0 and P1 is the same, as described above, the transmission efficiency is reduced to  $\frac{1}{3}$  compared to a case where no switching timing flag STF is added. On the other hand, in a case where the pulse cycle of each of the flag pulses P0 and P1 is a half of a pulse cycle T of the data D0 ( $T/2$ ), the transmission efficiency is only reduced to  $\frac{1}{2}$ . Thus, relatively speaking, the transmission efficiency is improved.

Variation Example

(258) Moreover, in this embodiment, a case where the number of transmission channels of the signal transmission device 400 is two is shown as an example, the number of transmission channels can be three or more. Degradation of the transmission efficiency can be further suppressed as the number of transmission channels increases.

(259) For example, in a case where a series of data included in each of the input pulse signals INA and INB is distributed to three insulated transmission paths and mixed data strings are generated, the transmission efficiency caused by the switching timing flag STF added is reduced to  $\frac{2}{3}$ .

Moreover, if the number of transmission channels is four, the transmission efficiency is reduced to

¾; if the number of transmission channels is five, the transmission efficiency is reduced to ⅔.  
(260) Moreover, the embodiment can be applied in a manner irrelevant to the type (coils, capacitors and optical components) of the insulated transmission path **420**.

(261) <Note>

(262) A summary of the various embodiments of the above description is given in the note below.

(263) For example, a signal transmission device disclosed in the present application is configured as (a first configuration), comprising: a transmitting circuit, configured to generate a plurality of mixed data strings from a plurality of input data strings; a plurality of insulated transmission paths, configured for insulation transmission of the plurality of mixed data strings; and a receiving circuit, configured to restore a plurality of output data strings corresponding to the plurality of input data strings from the plurality of mixed data strings that are insulated and transmitted via the plurality of insulated transmission paths.

(264) The signal transmission device of the first configuration can also be configured as (a second configuration), wherein the transmitting circuit is configured to generate the plurality of mixed data strings by sequentially switching data included in each of the plurality of input data strings and outputted to one of the plurality of insulated transmission paths.

(265) The signal transmission device of the second configuration can also be configured as (a third configuration), wherein the transmitting circuit is configured to add a switching timing flag to each of the data when generating the plurality of mixed data strings, and the receiving circuit is configured to restore the plurality of output data strings according to the switching timing flag.

(266) The signal transmission device of the third configuration can also be configured as (a fourth configuration), wherein the switching timing flag includes a pulse edge from a high level to a low level or from a low level to a high level.

(267) The signal transmission device of the fourth configuration can also be configured as (a fifth configuration), wherein the receiving circuit includes: an edge detection unit, configured to detect the pulse edge of the switching timing flag and generate an edge detection signal; a delay unit, configured to generate a delayed edge detection signal by applying a delay to the edge detection signal; a clock separation unit, configured to separate the delayed edge detection signal into a plurality of clock signals; a data import unit, configured to import data included in the plurality of mixed data strings into one of a plurality of separated data strings in synchronization with the plurality of clock signals; and an output timing adjustment unit, configured to adjust an output timing of each of the plurality of separated data strings in synchronization with the plurality of clock signals to generate the plurality of output data strings.

(268) The signal transmission device of any one of the third to fifth configurations can also be configured as (a sixth configuration), wherein the switching timing flag has a shorter cycle than a cycle of the data.

(269) The signal transmission device according to any one of the first to sixth configurations can also be configured as (a seventh configuration), wherein the data is one bit or multiple bits.

(270) The signal transmission device according to any one of the first to seventh configurations can also be configured as (an eighth configuration), wherein each of the plurality of insulated transmission paths includes a coil, a capacitor or an optical component as an insulated transmission element.

(271) Moreover, for example, an industrial apparatus disclosed in the present application is configured as (a ninth configuration), including the signal transmission device of any one of the first to eighth configurations.

(272) Moreover, for example, a vehicle disclosed in the present application is configured as (a tenth configuration), including the signal transmission device of any one of the first to eighth configurations.

OTHER

(273) Further, in addition to the embodiments, various modifications may be made to the technical

features disclosed by the present disclosure without departing from the scope of the technical inventive subject thereof. That is to say, all aspect described in the embodiments are illustrative rather than restrictive. Moreover, it is to be understood that the technical scope of the present disclosure is to be defined in accordance with the appended claims, and encompasses all equivalent meanings covered by the scope and all modifications made within the scope of the appended claims.

## Claims

1. A signal transmission device, comprising: a transmitting circuit configured to generate a plurality of mixed data strings from a plurality of input data strings; a plurality of insulated transmission paths configured for insulation transmission of the plurality of mixed data strings; and a receiving circuit configured to restore a plurality of output data strings corresponding to the plurality of input data strings from the plurality of mixed data strings that are insulated and transmitted via the plurality of insulated transmission paths.
  2. The signal transmission device of claim 1, wherein the transmitting circuit is configured to generate the plurality of mixed data strings by sequentially switching data included in each of the plurality of input data strings and outputted to one of the plurality of insulated transmission paths.
  3. The signal transmission device of claim 2, wherein the transmitting circuit is configured to add a switching timing flag to each of the input data strings when generating the plurality of mixed data strings, and the receiving circuit is configured to restore the plurality of output data strings according to the switching timing flag.
  4. The signal transmission device of claim 3, wherein the switching timing flag includes a pulse edge from a high level to a low level or from a low level to a high level.
  5. The signal transmission device of claim 4, wherein the receiving circuit includes: an edge detection unit configured to detect the pulse edge of the switching timing flag and generate an edge detection signal; a delay unit configured to generate a delayed edge detection signal by applying a delay to the edge detection signal; a clock separation unit configured to separate the delayed edge detection signal into a plurality of clock signals; a data import unit configured to import data included in the plurality of mixed data strings into one of a plurality of separated data strings in synchronization with the plurality of clock signals; and an output timing adjustment unit configured to adjust an output timing of each of the plurality of separated data strings in synchronization with the plurality of clock signals to generate the plurality of output data strings.
  6. The signal transmission device of claim 3, wherein the switching timing flag has a shorter cycle than a cycle of the input data strings.
  7. The signal transmission device of claim 1, wherein the input data strings are one bit or multiple bits.
  8. The signal transmission device of claim 1, wherein each of the plurality of insulated transmission paths includes a coil, a capacitor or an optical component as an insulated transmission element.
  9. An industrial apparatus, including the signal transmission device of claim 1.
  10. A vehicle, including the signal transmission device of claim 1.
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